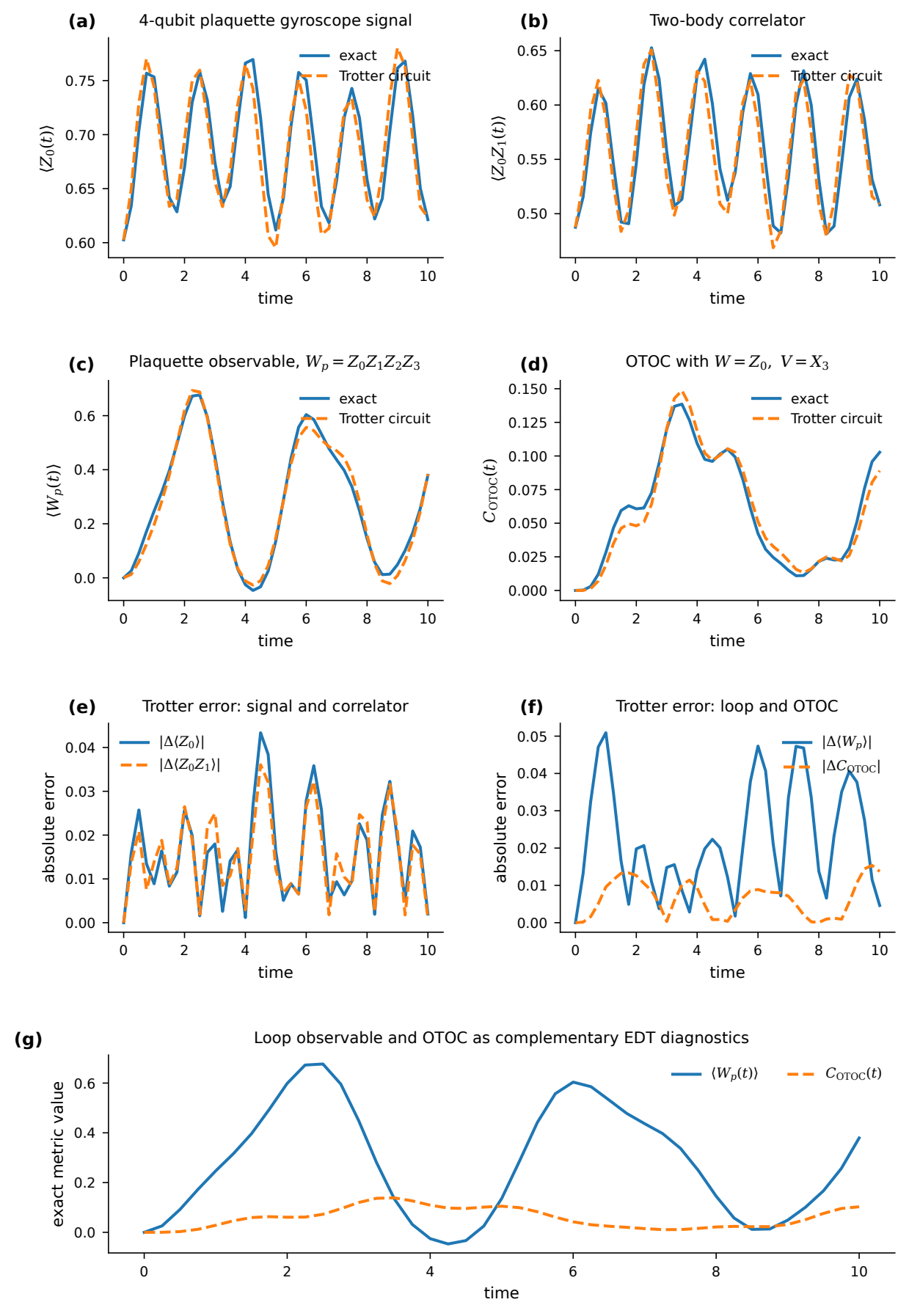
(a)
4-qubit plaquette gyroscope signal
exact
Trotter circuit
$\langle Z_0(t)\rangle$
time
(b)
Two-body correlator
exact
Trotter circuit
$\langle Z_0Z_1(t)\rangle$
time
(c)
Plaquette observable, $W_p = Z_0Z_1Z_2Z_3$
exact
Trotter circuit
$\langle W_p(t)\rangle$
time
(d)
OTOC with $W = Z_0,\ V = X_3$
exact
Trotter circuit
$C_{\mathrm{OTOC}}(t)$
time
(e)
Trotter error: signal and correlator
$|\Delta\langle Z_0\rangle|$
$|\Delta\langle Z_0Z_1\rangle|$
absolute error
time
(f)
Trotter error: loop and OTOC
$|\Delta\langle W_p\rangle|$
$|\Delta C_{\mathrm{OTOC}}|$
absolute error
time
(g)
Loop observable and OTOC as complementary EDT diagnostics
$\langle W_p(t)\rangle$
$C_{\mathrm{OTOC}}(t)$
exact metric value
time